



Modernizing the FBI's Criminal Justice Information Services (CJIS) Systems with Secure Cloud Innovation

How M&S Consulting Enabled Secure Cloud Transformation for the FBI's CJIS Division



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Executive Summary

The Federal Bureau of Investigation's Criminal Justice Information Services (CJIS) Division is the largest division within the FBI and serves as the technological foundation for law enforcement and national security agencies across the United States. CJIS systems—such as the National Crime Information Center (NCIC), Uniform Crime Reporting (UCR), National Data Exchange (N-DEx), and Next Generation Identification (NGI)—process millions of queries every day, underpinning nearly every criminal justice interaction nationwide.

However, much of this infrastructure was originally designed decades ago—based on mainframes and monolithic applications that struggled to keep pace with the FBI's growing mission and cybersecurity requirements. The modernization of these systems required a trusted technical partner with proven expertise in federal-scale data management, cybersecurity compliance, DevSecOps, and AWS cloud migration.

For over ten years, M&S Consulting has supported the CJIS Operations and Maintenance Professional Services (COMPS) II contract—delivering a complete suite of modernization services spanning cloud architecture, automation, cybersecurity hardening, continuous compliance, and agile transformation. Our work directly supported the FBI's ability to maintain 24/7 mission continuity, achieve and sustain Authorization to Operate (ATO) under evolving federal cybersecurity frameworks, and accelerate data availability for law enforcement agencies nationwide.

The result was a secure, cloud-enabled modernization of some of the most sensitive systems in government—accomplished without operational disruption and while strengthening the FBI's overall cyber posture.



The Challenge

Transforming Legacy Systems Without Compromising Mission Readiness

The CJIS environment comprises dozens of mission-critical systems that collectively form the backbone of U.S. criminal justice operations:

NCIC (National Crime Information Center) – Real-time law enforcement data clearinghouse with over 16 million active records.

N-DEx (National Data Exchange) – A cross-jurisdictional data platform enabling investigators to “connect the dots” across agency boundaries.

UCR (Uniform Crime Reporting) – The FBI’s statistical reporting system for crime data across 18,000+ agencies.

NGI (Next Generation Identification) – The world’s largest biometric repository, supporting fingerprints, palm prints, irises, and facial recognition.

LEEP (Law Enforcement Enterprise Portal) – The secure web gateway that connects all agencies to CJIS applications.

Each system operated on legacy, on-premises environments—some still linked to mainframe architectures—facing major modernization barriers:

Aging technology stacks (Oracle 11g/12c, RHEL 6, and legacy Java frameworks).

Manual operations and patching, increasing risk and downtime.

Siloed data models prevent integrated analytics across systems.

Escalating cybersecurity mandates requiring continuous vulnerability management and STIG compliance.

Rigid deployment cycles that slowed the rollout of new law enforcement capabilities.

CJIS needed a modernization approach that would preserve continuity of mission while migrating core workloads to the AWS GovCloud (US) and Microsoft Azure Government environments—both compliant with FedRAMP High and FBI CJIS Security Policy standards.

This challenge required not only deep cloud and DevSecOps expertise, but also operational empathy with an understanding that no downtime was acceptable in a 24/7 national security mission.



M&S Consulting's Role

Engineering the Future of Federal Justice Infrastructure

M&S Consulting was entrusted to deliver end-to-end modernization support under COMPS II—serving across multiple CJIS programs simultaneously and acting as the technical backbone of the transformation effort.

Our multidisciplinary team provided the following key capabilities:

Cloud Architecture and Migration (AWS GovCloud + Azure Government)

M&S provided the lead AWS Cloud Architect responsible for developing the initial cloud migration roadmap and executing proof-of-concept (PoC) validations for CJIS systems.

This work established the foundation for all subsequent cloud migrations across the CJIS portfolio.

KEY ACCOMPLISHMENTS

- Designed and validated AWS landing zones with segmented Virtual Private Clouds (VPCs), subnets, IAM controls, and network segmentation aligned to the CJIS Security Policy.
- Migrated mission-critical applications such as UCR, N-DEx, and Crime Data (CD)—from legacy Oracle environments to AWS GovCloud, leveraging:
 - Amazon EC2 for compute elasticity.
 - Amazon RDS & Aurora PostgreSQL for managed relational databases.
 - AWS Lambda and API Gateway for event-driven automation.
 - Amazon S3, DynamoDB, SQS, and SNS for high-throughput data ingestion and storage.
- Automated infrastructure provisioning and configuration management using Ansible, Jenkins, and Terraform-like IaC scripts.
- Supported cross-cloud operations by managing complementary Azure-hosted environments (including IBM Rational tools and virtual machine orchestration).

This multi-cloud strategy allowed CJIS to maintain operational redundancy while progressively retiring legacy on-premises infrastructure.



Cybersecurity and Continuous ATO Compliance

CJIS operates under some of the most stringent security requirements in the federal landscape. M&S implemented a multi-layered cybersecurity program integrating continuous monitoring, vulnerability remediation, and automated compliance enforcement.

KEY COMPONENTS INCLUDED

- Nessus Security Center Scans – Conducted every two weeks to detect vulnerabilities across Linux OS, Oracle, and application layers.
- OpenSCAP Scanning – Automated compliance verification aligned to the CJIS Security Policy and DISA STIGs.
- IBM BigFix Integration – Real-time visibility into patch levels and configuration drift.
- Splunk Enterprise Dashboards – Built by M&S to correlate security logs, system events, and failed login attempts in real-time.
- Jenkins + Ansible Pipelines – Automated patching cycles across RHEL servers using Red Hat Satellite Server.
- Plan of Action & Milestones (POA&M) tracking and exception management workflows** integrated with ISSO oversight.**

Through these measures, M&S not only helped CJIS achieve ATO on its new cloud platforms but also established a continuous compliance model, a first for several of the systems involved.

Our dashboards and automation reduced manual review effort by over 60% and accelerated remediation cycles, directly strengthening the FBI's cyber resilience.

Data Modernization and Big Data Integration

Modernizing the UCR program represented one of the largest federal data transformation initiatives in law enforcement history.

- Developed new relational data models spanning 300+ tables and twelve different form types, migrating over 180 million records from mainframe exports to Oracle 12.2 and PostgreSQL RDS environments.
- Automated publication data models (ROLAP)—over 50 complex BI views built to serve the Oracle Business Intelligence (OBIEE) platform.
- Automated data validation and report generation, replacing months of manual work with nightly, fully automated builds.
- Served as the data backbone for the Crime Data Explorer (CDE)—the FBI's public portal for national crime statistics.



This modernization allowed the FBI to decommission its legacy mainframe in 2018 and continue uninterrupted publication of national crime statistics—marking one of the most significant data migrations in FBI history.

DevSecOps and IT Service Management (ITSM) Enablement

M&S implemented a unified DevSecOps and ITSM model across CJIS programs using the Atlassian Suite (Jira, Confluence, Bitbucket, Bamboo) and Jira Service Management (JSM).

OUR CONTRIBUTIONS INCLUDED

- Building and maintaining Continuous Integration/Continuous Delivery (CI/CD) pipelines across value streams.
- Training and onboarding teams to JSM for ITSM compliance and service ticketing.
- Automating Atlassian upgrades and patches via scripting, reducing manual errors and downtime.
- Enabling integration between Atlassian and GitLab for end-to-end visibility across agile releases.

This transformation empowered CJIS teams to operate in a Scaled Agile Framework (SAFe) with two-week sprint cycles—a major cultural and operational evolution for a federal agency of this scale.

Mobile and Field Innovation: The Galton Application

M&S was instrumental in maintaining and expanding Galton, the FBI's mobile application suite supporting NGI Rap Back enrollment and notifications.

THE PLATFORM INCLUDES

- A secure web management portal,
- A desktop enrollment system, and
- An Android application for field use.

These tools allow law enforcement to receive real-time notifications when individuals under trust or supervision (teachers, federal employees, parolees) are linked to new criminal activity.

Since deployment, **Galton has processed over 8,000 Rap Back enrollments and 32,000 NGI responses**—making it a cornerstone of modern law enforcement biometrics.



Results and Impact

Through its decade-long engagement, M&S Consulting's work has become foundational to the FBI's modernization success.

Key results include:

OBJECTIVE	OUTCOME
Mainframe Decommissioning	Legacy UCR system fully retired; new UCR-TR system became an authoritative record in 2018.
Cybersecurity Posture	Continuous ATO maintained across multiple CJIS systems with automated monitoring and POA&M tracking.
Cloud Modernization	Multiple systems migrated to AWS GovCloud and Azure Government with zero operational disruption.
Automation & Efficiency	Patch and deployment cycles reduced from days to hours; security dashboards automated in Splunk.
Data Transparency	Real-time, publicly accessible crime data via the FBI's Crime Data Explorer.
Operational Agility	SAFe Agile model institutionalized across CJIS teams, with CI/CD fully embedded.

M&S's work enhanced national law enforcement coordination, reduced operational risk, and set a precedent for secure, scalable cloud adoption in federal agencies handling sensitive data.



Conclusion

The CJIS modernization effort underscores how strategic partnerships and technical expertise can redefine what’s possible in government IT transformation. M&S Consulting’s decade-long collaboration with the FBI demonstrates our ability to integrate deep technical capability with mission-critical reliability, delivering modernization that strengthens both operational effectiveness and public trust.

By combining AWS-native architectures, rigorous cybersecurity practices, and federal-scale automation, M&S helped the FBI establish a blueprint for the future of secure, cloud-enabled law enforcement systems—one that will continue to evolve as technology and national security needs advance.

Technologies and Tools

CATEGORY	TECHNOLOGIES & FRAMEWORK
Cloud Platforms	AWS GovCloud (EC2, Lambda, RDS, Aurora, S3, API Gateway, DynamoDB, SQS, SNS, IAM), Microsoft Azure Government
Automation & CI/CD	Jenkins, Ansible, GitLab, Red Hat Satellite, Terraform, Atlassian Bamboo
Cybersecurity & Compliance	Splunk, Nessus, BigFix, OpenSCAP, DISA STIGs, POA&M Tracking
Data & Analytics	Oracle 12.2, PostgreSQL RDS, OBIEE, ROLAP Models, Crime Data Explorer
Collaboration & ITSM	Atlassian Suite (Jira, Confluence, JSM, Bitbucket), SAFe Agile Framework
Field & Mobile	Galton NGI Applications (Web Portal, Desktop Client, Android App)



About M&S Consulting

M&S Consulting is an AWS Advanced Tier Partner and technology consulting firm specializing in cloud modernization, cybersecurity, DevSecOps, and data transformation. We partner with federal, state, and commercial clients to modernize mission systems, improve compliance, and drive measurable business outcomes.

Our team's work with the FBI CJIS Division exemplifies our mission: to engineer technology solutions that empower organizations to deliver their mission securely, efficiently, and with lasting impact.